SCIENCE 1.6 **Name:**_____

WORKSHEET FIVE **FORCES AND NEWTON'S LAWS**

1. (a) What is the SI unit for mass? ***Kilogram*** What is the symbol for this unit? ***kg***

   (b) Write a definition for mass. ***Mass is the amount of matter in an object***

   (c) Forces are sometimes described by the words **push, pull** and **twist.** Give an example of each of these three types of forces.

   Push ***kicking a ball***

   Pull ***catching a fish***

   Twist ***opening a screw-top lid***

2. List two advantages and two disadvantages of friction.

   Advantages ***We can grip things. We can slow down with brakes.***

   Disadvantages ***Wear and tear on moving parts. Energy is wasted as heat.***

3. Explain why a falling object will reach a constant speed of fall (Terminal velocity).

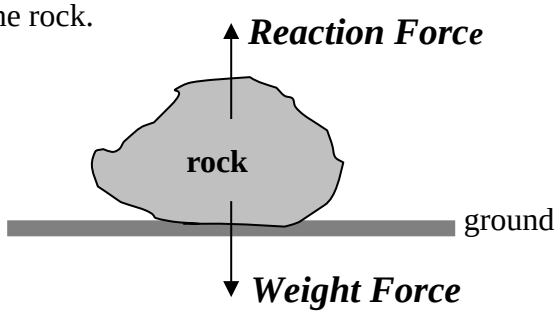
   As the object falls, the drag force (air resistance)e acting on it increases with speed. At the terminal velocity, the drag force is equal and opposite to the weight forceand the object stops accelerating.

4. The forwards in a Provincial rugby team have the following masses: **120kg, 110kg, 105kg 100kg, 98kg, 95kg, 90kg and 90kg**. Calculate the total mass of the pack and also what this forward pack would weigh in Newtons.

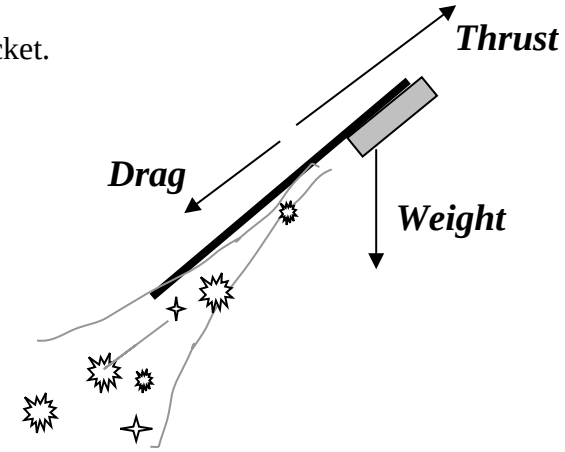
   Total mass = 808kg. The weight force on this = 808 x 10 = 8080 N

5. Write a definition for weight. ***Weight is the force of gravity on a mass.***

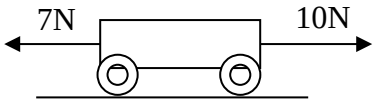
6. Draw and label all the forces acting on the rock.



7. Draw and label all the forces acting on the skyrocket.

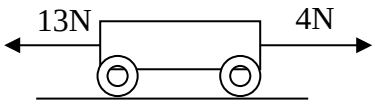


8. In the following diagrams calculate the net force on the 1 kg trolleys and also their acceleration.

(a) 

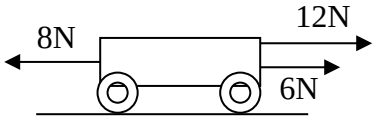
Nett Force = ***3 N to the right***

Acceleration = ***3 ms^{-2}***

(b) 

Nett Force = ***9 N to the left***

Acceleration = ***9 ms^{-2}***

(c) 

Nett Force = ***10 N to the right***

Acceleration = ***10 ms^{-2}***

9. **"An object will not accelerate when the forces acting on it are balanced**". Explain what this statement means.

When the forces acting on an object cancel each other out, the law F = ma tells us that the acceleration produced will be zero.

10. Use the equation ***F = ma*** to find the missing values in the chart below.

Force (N)	***300***	***50***	***4.5***	800	45	24	1000
Mass (kg)	100	2.5	0.3	80	5	***0.5***	***5***
Acceleration (ms^{-2})	3	20	15	***10***	***9***	48	200